ALAN TSENG

+886(988) 486-089 ◇ Taipei, Taiwan
alanhc.tseng1999@gmail.com ◇ linkedin.com/in/alanhc316 ◇ alanhc.github.io

EDUCATION

Master of Computer Science, National Taiwan University 2021-2024
GPA: 4.23/4.3

SELECTED EXPERIENCE

Software Engineer Intern Jul 2022 - Nov 2022
Line Corporation Taipei, Taiwan

- Developed and deployed a Swagger-based API development tool, streamlining API development by 20% and ensuring single source of truth. Seamless Kubernetes integration with optimized performance through DNS and load balancing configuration.
- Leveraged Nx to develop a scalable frontend architecture within a monorepo structure, and collaboration with the Korean backend team.
- Built a merchant management system during an internal hackathon using Next.js and Spring Boot, leveraging NFTs for innovative business empowerment.
- Presented a talk about Ethereum and Decentralized Application Development within an internal study group.

SELECTED PROJECTS

[**Research**] **FansTick** (Solidity, TypeScript, React Native, Python, IPFS, GunDB)

- TBD

[**Research**] **Human tremor symptoms detection** (Python, Tensorflow)

- Developed an LSTM-RNN model with Tensorflow Functional API for tremor frequency and noise detection.
- Achieved an average accuracy of 94.8% on a dataset of 160,000 simulated signals.

[**Course Project**] **Vaccine Tracking System** (Python, Solidity)

- Developed a blockchain-based vaccine tracking system utilizing Bloom Filter for efficient and rapid vaccine batch number verification with lower time complexity O(k) and space complexity O(m).

[**Course Project**] **Drum.io** (JavaScript, C, SocketIO, Arduino, Tensorflow.js)

- Designed and implemented a KNN-based multiplayer air drum game, awarded 1st place in Interactive Media Programming (52 participants) and 3rd place by collective vote (50 participants).
- Real-time analysis of acceleration data from an embedded device enabled intuitive air drumming and facilitated online competition.

[**Course Project**] **Create your dancing robot** (C, OpenGL)

- Got First place in the Computer graphics final project in a collective vote of 52 people.
- Implemented a model control system in OpenGL, enabling a robot to jump, inspired by Maya's graphical model control system.

[**Course Project**] **Landing Planet** (Java, Android)

- Received Second Place by collective vote (50 participants) for developing a space exploration game using Java and "Computer Graphics" concepts.

EXTRA-CURRICULAR ACTIVITIES

Developer Student Club (2019-2021) - Advisor by Yuh-Pyng, Shieh and Google Developers

- Selected as one of 800 leaders among over 3,000 passionate applicants worldwide for driving peer learning and community in a 2-year professional development program.
- Led and grew a technical student club, fostering a vibrant community and engaging over 73 attendees through talks on Deep Learning, PoseNet, Python & Colab, and Flutter.

ACM Competitive Programing Club (2017-2021) - Supervised by Jeng-Sheng Yeh

- Enhanced programming skills through practice and problem-solving on platforms like UVa Online Judge.

Volunteered at exhibitions

- Assisted with Developer Student Club (DSC) and Google Developer Groups (GDG) events at SITCON(Student Information Technology Conference) and COSCUP(Conference for Open Source Coders, Users and Promoters).
- Provided technical support at the ART FUTURE exhibition showcasing blockchain applications.

HONORS

Research Excellence Award

- Recognized for outstanding research contributions, placing among the top 1% of graduating students.

Intel Andy Grove Scholarship

- Selected as one of 367 recipients worldwide, out of over 2,000 applicants, for demonstrating academic excellence, leadership potential, and promising accomplishments.

College Student Research Scholarship

- Awarded a prestigious scholarship by the Ministry of Science and Technology, Taiwan, for exceptional research potential.

PUBLICATIONS

- Sung, H.-M., Chen, T., **Tseng, H.-C.**, Prayogo, B., Lin, J.-Y., & Hung, Y.-P. (2023). **akaTick: Hybrid Mobile E-Ticketing System Based on Non-Fungible Tokens**. 2023 IEEE International Conference on Metaverse Computing, Networking and Applications (MetaCom), 686–687.
- **Tseng, H.-C**., Tu, C.-W., Huan, X.-Y., & Chia, T.-L. (2020). **Nighttime vehicle light detection based on deep learning and image enhancement**. 2020 IPPR Conference on Computer Vision, Graphics, and Image Processing (CVGIP), 1–8.

SKILLS

- Blockchain(EVM Compatible Chains(L1/L2), Tezos, COSMOS, Solana), Artificial Intelligence(Machine Learning, DeepLearning), Web(Frontend/Backend) Development, Cloud-Native Applications, Computer Vision, Socket, Chrome Extension, SSG/SSR, MVC,
- Python, C/C++, JavaScript/TypeScript, Solidity, Java, SmartPy, Swift, Dart, C#, Matlab
- FastAPI/Flask, NodeJS, React(NextJS, React Native), Tensorflow, Scikit-Learn, OpenCV, OpenGL, Numpy, Pandas, Matplotlib, Processing(p5), Flutter
- NoSQL(MongoDB), SQL(PostgreSQL, MySQL), p2p(orbitDB, GunJS), IPFS